

Detection and quantification of integrated vector copy number by multiplex droplet digital PCR in dual-transduced CAR T cells

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The success of chimeric antigen receptor (CAR) T cell therapies in refractory hematologic malignancies has prompted investigation of their efficacy in solid tumors. AUTO6NG is a dual-transduced GD2-targeting CAR that encodes distinct modules designed to enhance T cell activity in relapsed/refractory neuroblastoma. The ability to detect and precisely quantify vector copy number (VCN) for each integrated vector is essential for assessing the effect of each module on T cell tumor infiltration, persistence, and clinical activity. Droplet digital PCR (ddPCR) enables accurate, sensitive, and absolute quantification of specific nucleic acid sequences. Compared to standard detection of two targets, multiplex ddPCR assays allow simultaneous detection of up to four targets by selective modulation of signal amplitude while retaining the ability to quantify the target. We have developed a multiplex assay based on the two-channel system for simultaneous detection and quantification of three targets in AUTO6NG CAR T cells. The assay was highly specific, sensitive, accurate, and reproducible across time and samples. No differences were observed in measuring VCN between standard duplex and multiplex assays. Our results demonstrate that ddPCR is an accurate and cost-effective method for simultaneous detection of multiple targets in genomic DNA derived from engineered CAR T cells.

INTRODUCTION

Adoptive T cell therapy involves the genetic reprogramming of cytotoxic T cells, through the expression of chimeric antigen receptors (CARs) on the cell surface, enabling them to recognize cell surface antigen(s) expressed on malignant cells. CARs are composed of an extracellular domain designed to bind a target with high specificity, without major histocompatibility complex restriction, and intracellular signaling domains that promote T cell activation.^{1,2} CAR T cell therapy has been most extensively evaluated in B cell malignancies. Encouraging results have been reported for patients with chronic lymphocytic leukemia and acute lymphoblastic leukemia using CD19-directed CAR T cells.^{3–6}

Neuroblastoma is a rare embryonal tumor of the sympathetic nervous system, which occurs almost exclusively in children.⁷ Promising cell- and antibody-based therapies targeting a tumor-associated antigen,

the disialoganglioside GD2 that is highly expressed in neuroblastoma, have significantly improved overall survival.^{8–12}

AUTO6 is an autologous T cell transduced with a gamma-retroviral vector encoding both an anti-GD2 CAR and the RQR8 suicide gene. The ongoing phase I clinical study of GD2-targeted CART for refractory/relapsed neuroblastoma (NCT02761915) shows activity against disseminated disease without inducing on target/off tumor toxicity.¹¹ To improve the efficacy of AUTO6 in neuroblastoma and other solid tumor indications, we designed AUTO6NG using two retroviral vectors that encode additional cell programming modules to enhance expansion, persistence, and tumor infiltration by counteracting potential immunosuppressive factors within the tumor microenvironment. An assessment of peripheral blood or tumor biopsies for the prevalence of CAR T cells harboring such modules can provide critical information about the mechanisms of clinical activity and resistance.

Analysis of CAR T cell persistence has been traditionally performed by quantitative real-time PCR amplifying integrated viral vector sequences in genomic DNA obtained from biospecimens. More recently, droplet digital PCR (ddPCR) has emerged as a more accurate and precise method for quantification of vector copy number (VCN) as a surrogate for CAR T cell persistence in clinical trials.^{13–19} The standard ddPCR assay is a duplex reaction that enables simultaneous detection of viral vector sequences as well as an endogenous sequence to quantify genomic DNA input.¹³ The higher order multiplexing ddPCR assay has recently been adapted to allow multiplexed detection of more than two targets in the same droplet by varying the concentrations of different primer-probes using the same fluorophore to modulate the endpoint droplet signal amplitude in the reaction.^{20–22}

Herein, we have applied multiplexed ddPCR for the detection and precise quantification of two integrated targets sequences as well as an endogenous target in dual-transduced AUTO6NG CAR T cells.

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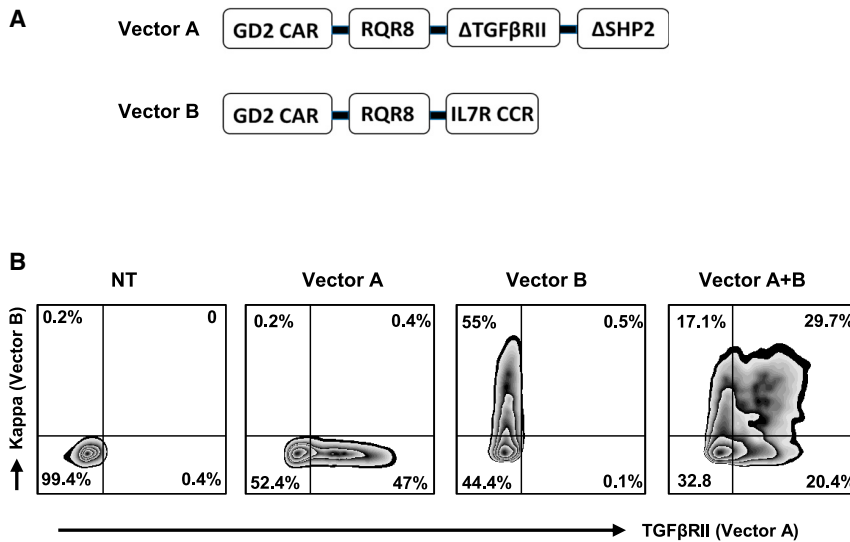


Figure 1. Viral vectors and generation of CAR T cells

(A) Schematic of the AUTO6NG CAR constructs. Vectors A and B consist of the following modules: GD2-targeting receptor (GD2 CAR), RQR8 suicide module, dominant-negative TGF β RII protein (Δ TGF β RII), truncated SHP2 protein (Δ SHP2), and the common γ chain (g-chain) heterodimerized with an IL7 receptor chimeric cytokine receptor (IL7R CCR) module. Sequences from SHP2 and g-chain were selected as targets to detect Vector A and Vector B, respectively. (B) AUTO6NG CAR T cells were transduced with two retroviral vectors to generate dual-transduced AUTO6NG and single-transduced control CAR T cells. Non-transduced (NT) T cells served as a negative control. The transduction efficiency was analyzed by flow cytometry using the CD34 antibody recognizing RQR8 construct that is coexpressed with CAR and the antibodies against TGF β RII and kappa light chain fused to the endodomain of the g-chain for Vector A and Vector B, respectively. The representative flow plots were gated on viable, single CD3+ T cells.

We provide evidence that the multiplex assay is as precise as the standard duplex assay. The assay is linear in a range of 5–100,000 copies per reaction for both vector sequences with no interference observed from peripheral blood genomic DNA ranging from 1.25 to 400 ng per reaction. Our results demonstrate that the multiplex assay is sensitive, precise, robust, and reproducible, highlighting its utility in characterization of integrated sequences in complex CAR T populations.

RESULTS

Generation of CAR T cells

Three types of CAR T cells were generated with the transduction of two retroviral vectors. Vector A single-transduced CAR T cells (Vector A) express a CAR against the GD2 target, a dominant-negative TGF β RII protein (Δ TGF β RII) to block signaling from TGF β , a truncated SHP2 protein (Δ SHP2) to block PD1 signaling, and a RQR8 suicide receptor as a safety switch to enable control of CAR T cell activity after infusion.^{23–26} Vector B single-transduced CAR T cells (Vector B) express the GD2 CAR, the common γ chain (g-chain) heterodimerized with a constitutively signaling IL7 cytokine receptor chimeric cytokine receptor (IL7R CCR) encoding sequence to enhance CAR T cell persistence, and the RQR8 receptor^{26–28} (Figure 1A). Activated T cells were transduced with Vector A, Vector B, or both, and the transduction efficiency was analyzed by flow cytometry using the antibodies against RQR8 expressed on both vectors, TGF β RII and kappa light chain fused to the endodomain of the g-chain for Vector A and B, respectively (Figure 1B). Non-transduced (NT) cells served as a negative control. The cell gating strategy was described as live cells > single cells > CD3+ > CD34+ > TGF β RII (anti-TGF β RII) vs. g-chain (anti-Kappa). The cells transduced with Vector A (Vector A) showed 66% transduction efficiency based on CD34 expression with 75% positive staining for TGF β RII and negative staining for g-chain. Vector B single-transduced cells showed 61% transduction efficiency with 67% positive for g-chain and negative for TGF β RII. Dual-transduced AUTO6NG CAR T cells (Vector A + B) showed

70% positive staining for RQR8 with 19% TGF β RII single-positive, 21% g-chain single-positive, and 33% double-positive transduction. These data demonstrate the specificity of single- and dual-transduced CAR T cells as controls in multiplexed ddPCR assays.

Selection of primers and probes for each vector of AUTO6NG CAR T cells

To quantify the abundance of target vector sequences in genomic DNA derived from peripheral blood, we sought to identify unique sequences from Vector A and Vector B for amplification, aiming to maximize specificity, signal amplitude, and similar annealing temperatures. We selected sequences encoding dominant-negative SHP2 (SHP2) from Vector A and sequences encoding the g-chain heterodimerized with IL7R CCR for Vector B (Figure 1A). Sequences for both targets are codon-optimized and do not share homology with endogenous sequences present in the human genome. Three primer-probe pairs were designed for each target in FAM (6-carboxyfluorescein) channel, SHP2 for Vector A, and g-chain for vector B (Table 1). VCN in CAR T clinical trials is commonly expressed as the number of target sequences detected per microgram of genomic DNA. While each PCR reaction is loaded with 100–200 ng of genomic DNA as measured by spectrophotometry, quantification of an endogenous reference target by ddPCR is used to obtain a measure of the number of diploid genomes and, by inference, the precise quantity of gDNA in the PCR reaction. Commercially available primer/probe sequences from human ribonuclease P protein subunit p30 (RPP30) were used for this purpose (Table 1).

A ddPCR one-dimensional (1D) plot showed that SHP2-positive droplets were visible in the FAM channel with all three primer-probe pairs in single- (Vector A) and double-transduced cells (Vector A + B). There was no false positive signal detected in vector B or NT cells using Vector A reagents (Figure S2A). Similarly, g-chain was specifically detected in the FAM channel with all primer-probe pairs in

Table 1. Design of primer-probe sequences for SHP2, g-chain (FAM), and RPP30 (HEX)

	Forward primer (500 nM)	Reverse primer (500 nM)	Probe (250 nM)
SHP2-1	CAATGACGGCAAGTCCAAAG	GTCAGGCTGTGCAATCTCTC	ATGATCCGGTGCCAGGAAGTGAAG
SHP2-2	GACCCACATCAAGATCCAGAA	CCGTGGTGTTCATGTAGTA	CACCGGCGACTACTACGACCTGTA
SHP2-3	CAAAGTGACCCACGTGATGA	GGGTTTCCACCATAGGGTTT	ACGTCGTACTTCAGTTCCTGGCAC
g-chain-1	TGACTACTCCGAGAGACTGTG	ATAAGGGCTGTGCTGGTTG	CCTGGTGTCCGAGATCCCTCCTAA
g-chain-2	CCTGCTGTGCGTGTACTT	GAGAAAGTGCCGTGGTACTC	CTGGAACGACCATGCCAGAATC
g-chain-3	GGCAACTTCTCTGCTTGA	TTAGGAGGGATCTCGACAC	TGACTACTCCGAGAGACTGTGCCT
RPP30	AGTAACTTGTAAGTGGTAGTCATAGA	ATGTCAAGAGTAGGAGGACATTG	TCAGGCAGACTGACACTAGAGTTC

Vector B single- (Vector B) and double-transduced cells and no signal was detected in Vector A or NT cells (Figure S2B). RPP30 signal was detected in the HEX (hexachloro-fluorescein) channel. The VCN quantification for Vector A using SHP2 as the target with three pairs of primer-probe showed no difference in single- or double-transduced cells (Figure S2C). Based on the standard deviation of the replicates, positive signal intensity, and similarities in T_m , SHP2 pair-1 was selected as the target for Vector A. Similarly, all primer-probe pairs of g-chain showed precise and consistent VCN quantification, and g-chain pair-1 was selected as the Vector B target. Altogether, these data confirm that the designed primers conferred adequate specificity.

Establishment of the multiplex assay

The standard duplex assay enables detection of endpoint fluorescence amplitude and quantification of SHP2 and RPP30 or g-chain and RPP30 in the FAM and HEX channels, respectively, within a single reaction. Higher order multiplexing can be achieved by modulating the primer-probe concentrations for each target and deconvoluting the resulting distribution of signal on the 2D scatterplot. This enables detection of multiple targets simultaneously in two discrete optical channels while retaining the ability to quantify each target with precision. The primer-probe concentrations used for each target are listed in Table 2.

To establish the accuracy of the multiplex assay, quantification of VCN using the same gDNA extracted from dual-transduced AUTO6NG cells (Vector A + B) (Figure 1B) by ddPCR standard and multiplex assays was compared. The standard duplex analysis is illustrated by two-dimensional (2D) scatter graphs (Figures 2A and 2B). The thresholds were set up at 5,000 on the Y axis for SHP2 and g-chain (FAM) and at 3,000 on the X axis for RPP30 (HEX) to separate the positive and negative partitions. The black cluster in the lower left marks the dual-negative droplets. For the multiplex assay, both SHP2 and g-chain targets were detected in FAM channel, and RPP30 was detected in HEX channel. However, SHP2 and g-chain had relative concentrations of 200% and 80% of RPP30, respectively, (Table 2) resulting in eight different clusters (Figure 2C). The concentration of each target and thresholds used were the same as the standard duplex assays. We observed no significant difference between the duplex and the multiplex assays in terms of target copy

number per reaction or VCN per cell (Figures 2D and 2E), suggesting that one multiplex assay is as accurate as two standard duplex assays.

Determination of linearity, precision, limits of detection, and quantitation

The limit of detection (LoD) is the lowest target copy number in a PCR reaction that can be detected at a specified level of confidence, and the limit of quantification (LoQ) is the lowest copy number in a reaction that can be reliably quantified with an acceptable level of precision and accuracy.^{29,30} To establish the sensitivity and accuracy of the multiplex ddPCR system, the linear range, the LoD, and LoQ of the assay were determined.

To determine the linear range of SHP2 and g-chain VCN by multiplex assay, an 11-point dilution series (5–500,000 copies) of Vector A plasmid encoding SHP2 and Vector B plasmid encoding g-chain was prepared in a constant background of 100 ng human gDNA per reaction. No DNA template served as a negative control (Blank). Serial dilutions (10 replicates each) were then analyzed by ddPCR and compared with expected counts (Table S1). The QuantaSoft software failed to report measured values at the highest concentration of 500,000 copies per reaction; therefore, 100,000 copies per reaction was considered the upper LoD and quantitation for the assay. The mean, standard deviation (SD), and the coefficient of variation (CV) were calculated. Both SHP2 and g-chain showed good correlation between expected and measured values with R^2 values of 0.9985 and 0.9994, respectively, establishing 5–100,000 copies as the linear range of the assay (Figure 3A). The SHP2 and g-chain sequences in the reaction had no impact on the detected RPP30 copy number (Table S2).

We next calculated the lower LoD as the mean of Blank + (3.3 x SD of the lowest concentration where CV is less than 40%). The LoQ was calculated as the mean of Blank + (10 x SD of the lowest concentration where CV is less than 40%) (29). The LoD and LoQ for SHP2 target were calculated as 10 copies and 30 copies per reaction, respectively (Table S1). Similarly, the LoD and LoQ for g-chain target were calculated as 7 copies and 20 copies per reaction, respectively (Table S1).

Finally, we evaluated the effect of different quantities of gDNA extracted from peripheral blood on VCN for each target. A 12-point

Table 2. Primer-probe concentrations use for the duplex and multiplex assays

	Forward primer	Reverse primer	Probe
SHP2	1,000 nM	1,000 nM	500 nM
g-chain	400 nM	400 nM	200 nM
RPP30	500 nM	500 nM	250 nM

serial dilution of gDNA (10 replicates each) ranging from 400 ng to 0.01 ng per reaction was evaluated in a constant background of Vector A and Vector B sequences (Table S2). Expected and measured RPP30 copy numbers are strongly correlated with an R^2 value of 0.9896 (Figure 3B). Importantly, we observed no significant interference from 0.25–400 ng purified peripheral blood gDNA on SHP2 and g-chain VCN in this assay.

Assessment of variation in VCN quantification across time

To evaluate variation in quantifying VCN across time in the multiplex assay, we evaluated the VCN from three aliquots of the same batch of AUTO6NG CAR T cells cryopreserved and thawed at different time points (R1, week 0; R2, week 2; and R3, week 4). The VCN was established by multiplex assay in parallel with the standard duplex assay. We observed no difference in SHP2 or g-chain VCN between these two assays or across time points over a 2-week period (Figure 4). These results suggest that multiplexed ddPCR can be used to determine the VCN for each vector of dual-transduced AUTO6NG CAR T cells accurately and consistently across time.

DISCUSSION

ddPCR has several advantages compared to standard real-time PCR, including higher sensitivity and accuracy.¹³ It provides absolute quantification without external calibration curves, which reduces variability between assays and difficulty in measuring targets from samples with low numbers of cells. Standard duplex ddPCR is increasingly used for precise determination of persistence in CAR T clinical trials.^{14–19} Here, we have extended the ddPCR assay to detect two integrated viral sequences along with an endogenous sequence in the same droplet. To our knowledge, the present manuscript is the first study demonstrating multiplexed assessment of VCN in dual-transduced CAR T cells. Our results show that the multiplex assay is as sensitive, precise, and robust as the corresponding duplex analysis. However, the multiplexed assay described in this study does not provide single-cell-level information, but rather it quantifies proportional representation of each sequence at the population level including all single- and dual-transduced cells as well all NT nucleated cells from which gDNA is derived.

Accurate estimation of VCN in transduced cells is essential for monitoring CAR T cell expansion and long-term persistence after infusion in the treated patients.³¹ As engineered CAR T cells gain additional functionality to overcome immune challenges in the solid tumor setting, precise characterization of complex populations derived from biospecimens can lead to a better understanding of the interplay

between the tumor microenvironment and CAR T cell infiltration, expansion, and persistence. One powerful application of the multiplexed assay is to evaluate the relative abundance of specific modules in various compartments of a tumor biopsy to better understand immunosuppressive mechanisms and to guide the design of more effective vectors.

The multiplex assay provides a precise and reliable for rapid quantitation of the relative abundance of specific viral sequences in engineered T cells. It minimizes the cost for analysis and enables more information compared to the standard assay, particularly for the samples with limited input. Furthermore, the assay can be rapidly adapted to detect new target genomic DNA sequences, or transcripts when combined with a reverse transcription step. However, the reproducibility of the assay, such as the variation in measuring VCN across analysts, laboratories, and instruments is required for further evaluation.^{21,22}

In this study, we describe a triplex ddPCR assay that enables quantification of two target sequences through signal deconvolution. However, the platform is amenable to higher order multiplexing by addition of a second target to the HEX channel or through further refinements in reagent concentrations to modulate signal amplitude and introduction of automated signal deconvolution algorithms. Overall, multiplexed ddPCR offers a robust and cost-effective approach to rapid analysis of complex CAR T cell populations in clinical samples.

MATERIALS AND METHODS

AUTO6NG CAR T cell production

The AUTO6NG CAR T cell production method was prepared as described previously.¹¹ Briefly, peripheral blood mononuclear cells from healthy donors were activated with anti-CD3 (clone OKT3, 100 ng/mL, Miltenyi Biotec, Bergisch Gladbach, Germany) and anti-CD28 monoclonal antibodies (clone 15E8, 100 ng/mL, Miltenyi Biotec) and human recombinant human IL-7/IL-15 cytokines (10 ng/mL, PeproTech, London, UK) to induce T cell proliferation. The cells were then transduced with two retroviral vectors: Vector A encoded a GD2 CAR and SHP2/TGF β R2 dominant-negative modules, and Vector B encoded a GD2 CAR and IL7R CCR module. The cells were expanded for 5 days after initial transduction and were assessed by flow cytometry for transduction efficiency and by ddPCR analysis for VCN.

DNA isolation and quantification

Genomic DNA (gDNA) was extracted from AUTO6NG CAR T cells and NT control cells using a Qiagen QIAamp DSP DNA Blood Mini Kit (Qiagen, Hilden, Germany) according to the manufacturer's instructions. The purity and concentration of the isolated dnaA260/A280 ratio was measured using a Nanodrop One spectrophotometer (Thermo Fisher Scientific, Waltham, MA).

ddPCR duplex and multiplex assays

ddPCR duplex and multiplex assays were performed using a Bio-Rad Auto DG QX200 ddPCR system (Bio-Rad Laboratories, Hercules,

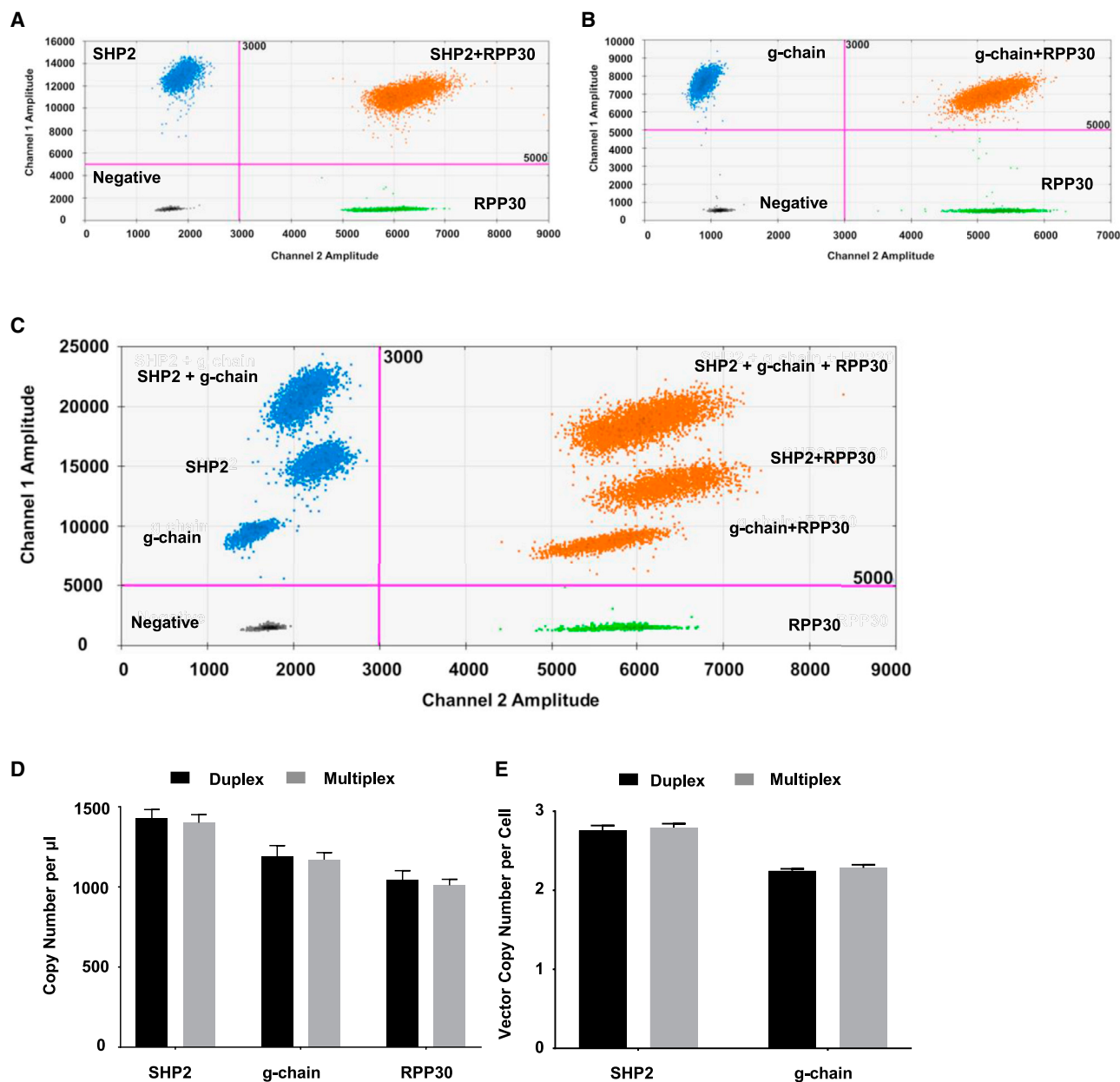


Figure 2. Comparison of SHP2 and g-chain VCN quantification by duplex and multiplex assays

ddPCR VCN analysis using the same gDNA extracted from AUTO6NG dual-transduced cells by standard and multiplex assays. Representative 2D scatterplots showing the detection of (A) SHP2/RPP30 and (B) g-chain/RPP30 positive droplets by a standard duplex ddPCR assay: double-negative (black), SHP2/g-chain single-positive (FAM: blue), RPP30 single-positive (HEX: green), and double positives (orange). (C) Representative 2D scatterplot showing SHP2 and g-chain positive droplets (FAM) and RPP30 (HEX) droplets in a multiplexed ddPCR reaction resulting in eight clusters. (D) Comparison of SHP2, g-chain, and RPP30 sequence copy numbers between the duplex and multiplex ddPCR assays. (E) Comparison of SHP2 and g-chain VCN assessments between duplex and multiplexed assays normalized to RPP30 (copies per cell). Bars indicate mean values with SDs ($n = 4$).

CA) according to the manufacturer's instructions. All designed pairs of primer-probe for targets were cross-checked with binding sites against the human genome to ensure target specificity (Primer-BLAST, NCBI) and purchased from IDT (Integrated DNA Technologies, Coralville, USA.). The details for primer-probe pairs are pro-

vided in Table 1. A temperature gradient analysis determined 59°C to be the optimal annealing temperature in the duplex and multiplex assays with SHP2 and g-chain vector constructs (Figure S1). ddPCR was performed using 1X ddPCR Supermix for probes (no dUTP, Bio-Rad Laboratories) and 20 units of HindIII restriction enzyme

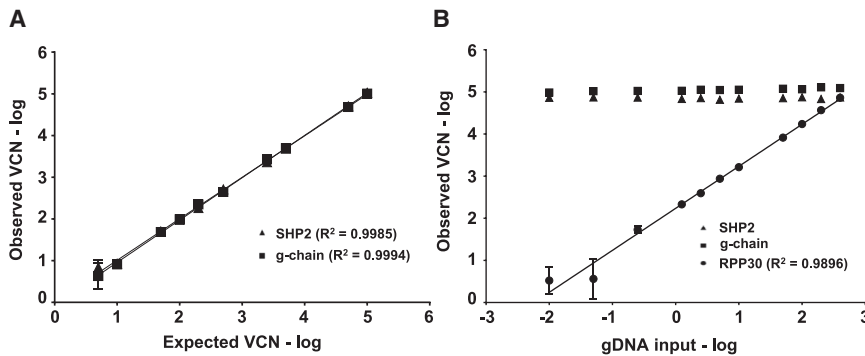


Figure 3. Determination of linearity, precision, and gDNA interference in the multiplexed ddPCR assay

(A) Detection of SHP2 and g-chain in a constant background of 100 ng gDNA (10-point serial dilutions from 100,000 to 5 copies, 10 replicates each). The expected and observed VCNs are indicated on the X and Y axes, respectively (Log10). Both SHP2 ($R^2 = 0.9985$) and g-chain ($R^2 = 0.9994$) show a high correlation between observed and expected values. (B) RPP30 assay linearity and interference with SHP2 and g-chain VCN assessment. RPP30 copy number was assessed using gDNA input ranging from 0.01 to 400 ng in a background of 71,614 and 112,307 copies of Vector A and Vector B per reaction, respectively.

(New England BioLab, Hitchin, UK) in a final 22 μ L per reaction with 100 ng of gDNA extracted from the cells. The concentrations of primer-probe were used as indicated. Following droplet generation, between 11,000 and 15,000 on average, the droplets were amplified using a C1000 Touch thermocycler (Bio-Rad Laboratories) with a 105°C heated lid. The thermal cycling program included the enzyme activation at 95°C for 10 min, 40 cycles of denaturing at 94°C for 30 s, and annealing at 59°C for 1 min at a ramp rate of 2°C/s, followed by enzyme deactivation at 98°C for 1 min. The droplets were held at 4°C prior to analysis. No template control reactions were performed using nuclease-free water. The plates containing the droplets were read with QX200 droplet reader (Bio-Rad Laboratories) using a two-channel setting to detect FAM and HEX. The positive droplets containing amplification targets were discriminated from negative droplets without amplification products by applying a fluorescence amplitude threshold. The threshold was set manually as indicated in the middle between the clusters using the 1D or 2D amplitude chart. The concentration of target amplicon per unit volume and estimation of VCN of target gene normalized to the reference gene (RPP30) were determined by QuantaSoft software version 1.7.4.0917 (Bio-Rad Laboratories, Hercules, CA). All the assays worked well according to the minimal information for publication of quantitative digital PCR experiments guidelines.^{21,22}

Flow cytometry analysis

The AUTO6NG CAR T cells were analyzed by flow cytometry to assess the transduction efficiency using fluorochrome-labelled antibodies including BUV395 anti-CD3 (clone UCHT1, 1:100 dilution, BD Biosciences, San Jose, CA), APC anti-CD34 (clone # QBEnd10, 1:50 dilution, R&D Systems, Abingdon, UK), FITC anti-TGF β RII (clone # 25508, 1:100 dilution, R&D Systems), PE anti-Kappa for g-chain (clone A8B5, 1:100 dilution, Thermo Fisher Scientific, Waltham, MA), and viability dye FVS780 (1:1,000 dilution, BD Biosciences). Samples were stained and acquired with BD Fortessa flow cytometer (BD Biosciences, San Jose, CA). Data were analyzed with FlowJo software (BD Biosciences, San Jose, CA).

Statistical analysis

GraphPad Prism version 7 software (GraphPad Software, La Jolla, CA) was used for statistical analysis. Data are presented as mean

with SD. Two-way ANOVA analysis was used for multiple group comparison. Two-tailed Pearson statistics with a confidence interval of 95% were used to determine correlation coefficients. Probability values (p) < 0.05 were considered statistically significant.

DATA AND CODE AVAILABILITY

We confirm that the data supporting the findings of the work are available within the article and the [supplemental information](#).

SUPPLEMENTAL INFORMATION

Supplemental information can be found online at <https://doi.org/10.1016/j.omtm.2023.07.003>.

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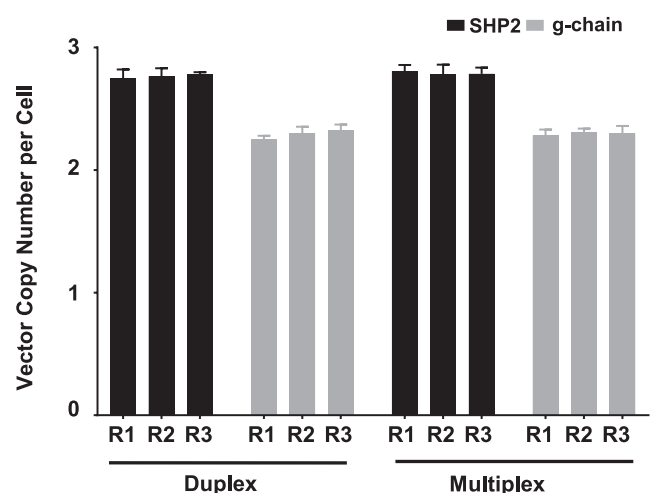


Figure 4. Assessment of short-term stability

Comparison of SHP2 and g-chain VCN assessment by duplex and multiplex ddPCR assays using cryopreserved CAR T cells across three time points spaced 2 weeks apart (R1, week 0; R2, week 2; and R3, week 4). The quantitation was established on VCN per cell. Bars indicate mean values with SDs ($n = 3$).

AUTHOR CONTRIBUTIONS

W.W., M.A.-H., and A.S.A. conceived of the presented idea. W.W. designed and performed the experiments. A.S.A. and M.A.-H. supervised the work and verified the analytical methods. W.W. and A.S.A. wrote the manuscript with support from M.A.-H.

DECLARATION OF INTERESTS

The authors declare no potential conflicts of interest.

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Supplemental information

**Detection and quantification of integrated
vector copy number by multiplex droplet
digital PCR in dual-transduced CAR T cells**

Wei Wang, Muhammad Al-Hajj, and Alireza S. Alavi

Supplemental Figures

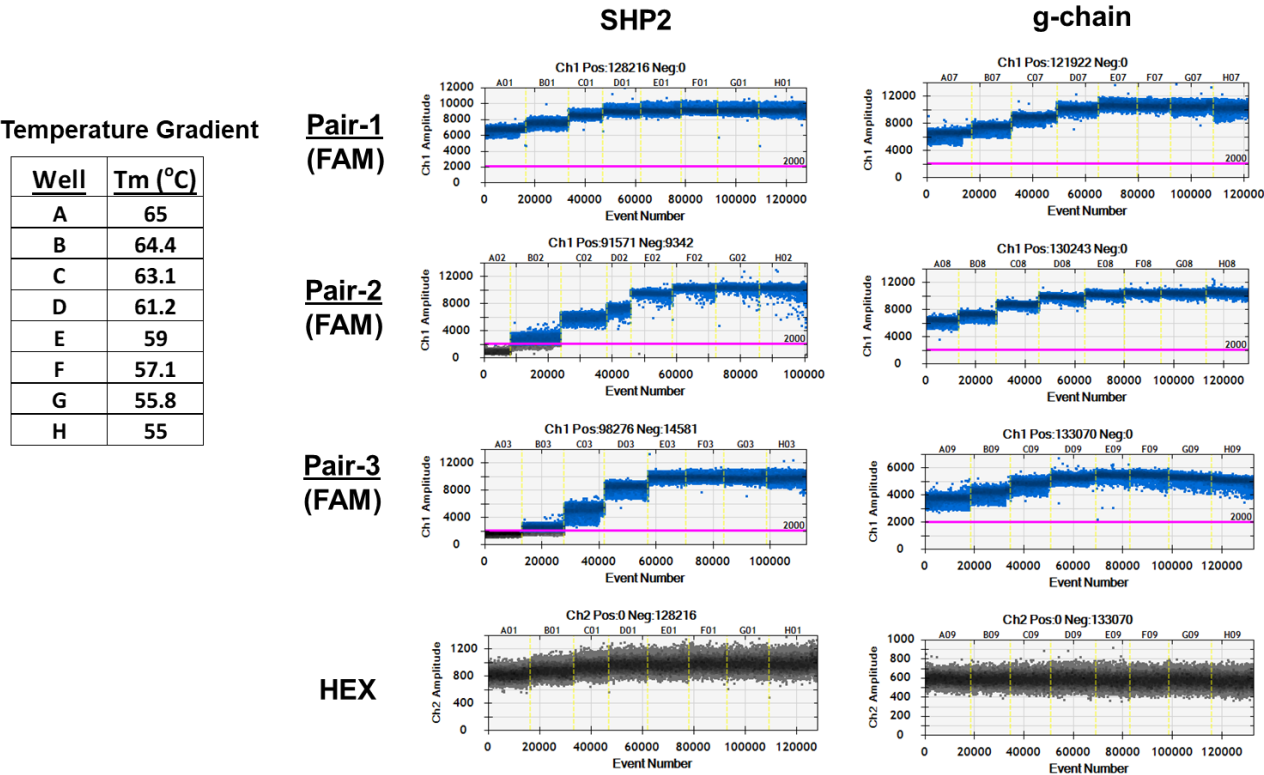


Figure S1. Determination of the optimal annealing temperature for ddPCR assay.

The annealing temperature (T_m) was optimized using a thermal gradient (55°C - 65°C) with all three pairs of primers-probe for each target (SHP2 and g-chain) at the same final concentration. 59 °C was selected as the optimal annealing temperature based on the highest temperature that produced the best separation of negative and positive droplets in FAM channel.

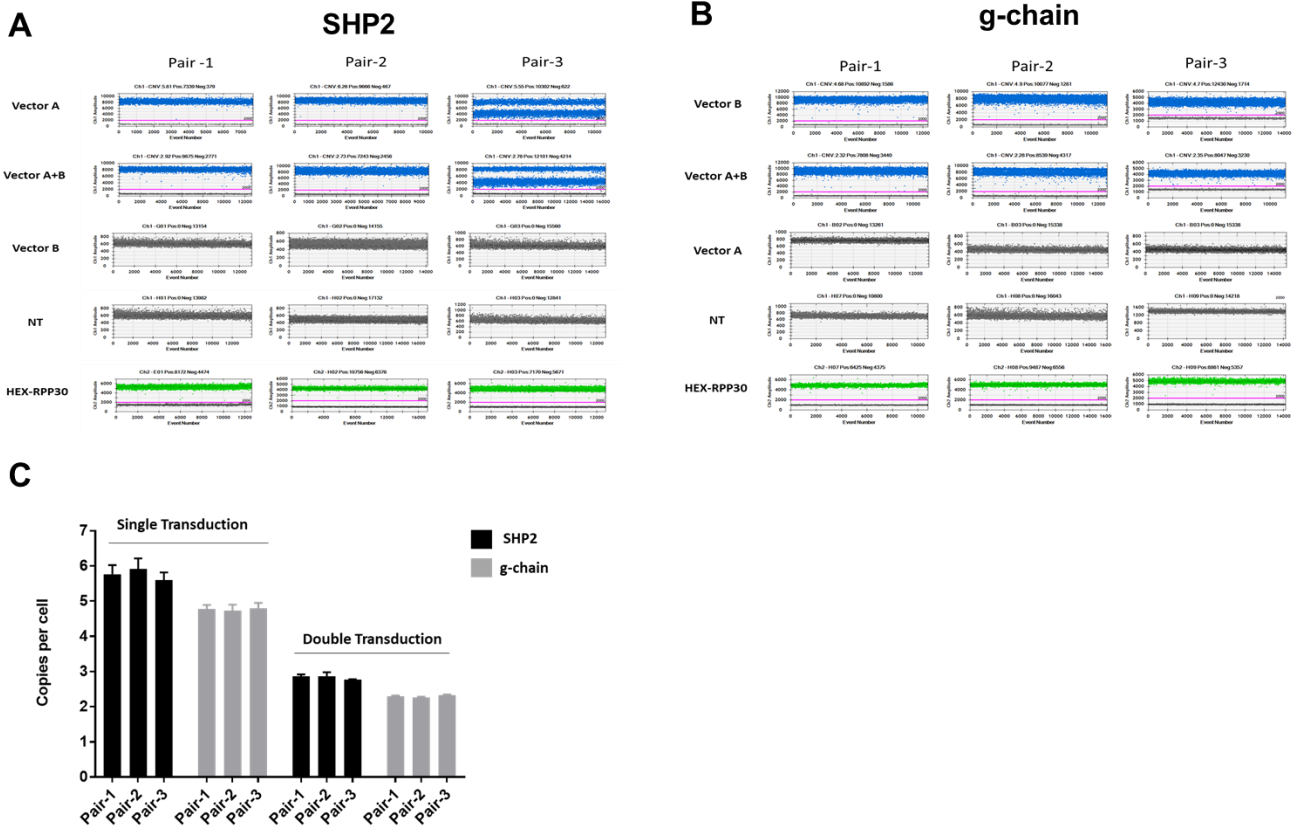


Figure S2. Analysis of primer-probes for SHP2 and g-chain.

Three specifically designed pairs of primers-probe were used to detect and quantify SHP2 (A) or g-chain (B) in FAM channel using Vector A single-transduced cells, Vector B single-transduced cells, Vector A+B double-transduced cells, and non-transduced cells (NT). RPP30 was measured in the HEX channel. (C) Comparison of SHP2 and g-chain VCN (copies per cell) quantification using each of the three pairs of primer-probe. Bars indicate mean values with SDs (n=3).

Supplemental Tables

Table S1. Determination of assay linear range, precision, LoD, LoQ.

RPP30													
gDNA (ng)	Rep1	Rep2	Rep3	Rep4	Rep5	Rep6	Rep7	Rep8	Rep9	Rep10	Mean	SD	CV%
100	19918	18423	17963	18170	18929	18584	18354	17756	18814	18538	18545	602	3.2
100	20861	20677	19941	19596	18377	20309	19159	19251	18354	18814	19534	901	4.6
100	21183	20792	20585	20033	21459	19458	20033	20102	19918	19274	20284	710	3.5
100	22011	21827	19412	20769	20286	20286	19596	19435	19274	18768	20166	1092	5.4
100	21781	20125	21160	20654	19803	20999	20033	19826	19964	19688	20403	707	3.5
100	20631	20930	22425	21206	20677	21275	20401	19872	20424	21068	20891	687	3.3
100	20792	21022	22264	23851	22080	20585	20240	19412	21068	19918	21123	1301	6.2
100	22126	21137	21643	20815	21137	20815	19389	19228	19274	19826	20539	1041	5.1
100	23598	22356	22402	22057	21413	20424	20792	20723	20378	19964	21411	1158	5.4
100	22356	20217	20861	22471	22586	21965	20976	20953	20654	20378	21342	908	4.2
100	22839	21206	22632	18975	21942	21597	20240	20424	20838	21114	21181	1157	5.5
0	0	0	0	0	0	0	0	0	0	0	0	0	N/A
SHP2													
Expected	Rep1	Rep2	Rep3	Rep4	Rep5	Rep6	Rep7	Rep8	Rep9	Rep10	Mean	SD	CV%
500000	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
100000	99560	108680	109060	110580	104500	110200	107540	104120	101840	104880	106096	3701	3.5
50000	52866	51414	51150	51678	49698	48609	50325	49599	48444	47388	50117	1692	3.4
5000	5313	5175	4761	4830	4830	4807	4646	4554	5014	4738	4867	235	4.8
2500	2254	2323	2323	2323	2484	2346	2139	2093	2208	2438	2293	123	5.4
500	580	499	561	501	543	541	531	462	455	474	515	43	8.3
200	202	186	156	207	193	154	191	173	179	189	183	18	9.7
100	92	76	109	99	102	89	96	89	106	102	96	9	10.2
50	60	55	44	51	41	58	58	39	53	48	51	7	14.4
10	10	9	9	9	10	6	11	9	7	8	9	1	15.4
5	6	9	8	6	11	12	8	5	4	4	7	3	39.4
0	0	0	0	0	0	0	0	0	0	0	0	0	N/A
g-chain													
Expected	Rep1	Rep2	Rep3	Rep4	Rep5	Rep6	Rep7	Rep8	Rep9	Rep10	Mean	SD	CV%
500000	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
100000	99590	97290	100510	105570	105110	105110	106260	103500	96140	97980	101706	3835	3.8
50000	52050	46950	47400	48900	51750	48000	46350	46500	48150	47700	48375	2012	4.2
5000	4644	6264	4230	4716	4806	5454	4158	4734	5688	5364	5006	669	13.4
2500	2277	2783	2875	2737	2990	2990	2369	2760	2668	2484	2693	246	9.1
500	580	375	382	439	426	391	506	414	455	497	446	65	14.5
200	186	258	235	253	225	230	175	193	260	244	226	31	13.7
100	86	92	109	76	119	119	109	106	92	86	99	15	15.0
50	64	58	48	39	53	48	46	51	44	41	49	8	15.5
10	7	12	7	12	9	8	10	6	6	8	8	2	23.8
5	2	6	2	6	11	2	4	13	3	5	5	4	71.4
0	0	0	0	0	0	0	0	0	0	0	0	0	NA

Table S2. Analysis of gDNA interference and dynamic range.

RPP30													
gDNA (ng)	Rep1	Rep2	Rep3	Rep4	Rep5	Rep6	Rep7	Rep8	Rep9	Rep10	Mean	SD	CV%
400	74400	78000	73440	73680	72480	72000	69120	70800	72240	73440	72960	2346	3.2
200	39840	38400	36768	38664	34248	35640	36480	34584	36432	37440	36850	1785	4.8
100	18336	17136	17496	16704	18672	17904	16608	15984	17232	16752	17282	834	4.8
50	8400	8376	8520	8424	9096	8160	7920	7584	7872	8064	8242	421	5.1
10	1716	1630	1906	1694	1615	1620	1474	1450	1603	1670	1638	128	7.8
5	785	1008	986	734	878	838	838	830	859	974	873	90	10.3
2.5	463	362	331	394	422	362	410	437	374	403	396	40	10.0
1.25	250	218	221	173	254	221	175	204	211	230	216	27	12.5
0.25	41	72	55	53	38	58	50	48	67	62	54	11	19.8
0.05	7	0	5	5	5	2	4	10	3	9	5	3	60.5
0.01	0	0	9	3	0	0	0	4	0	1	2	3	172.5
0	0	0	0	0	0	0	0	0	0	0	0	0	N/A
SHP2													
gDNA (ng)	Rep1	Rep2	Rep3	Rep4	Rep5	Rep6	Rep7	Rep8	Rep9	Rep10	Mean	SD	CV%
400	77040	78720	72960	76320	74160	73920	69600	71520	72240	72720	73920	2753	3.7
200	72000	67440	66000	70560	64080	64560	65280	63600	72240	73440	67920	3776	5.6
100	77760	72720	77040	73680	80640	76320	71520	67920	73440	76320	74736	3621	4.8
50	72960	72000	71520	75360	73680	72480	66960	62496	73920	71040	71242	3802	5.3
10	68400	70320	77040	71280	65520	66240	63120	67200	72240	75840	69720	4492	6.4
5	58152	69120	67680	62640	68880	64320	63360	55728	71040	73440	65436	5643	8.6
2.5	70080	63840	63120	68880	68160	70080	81120	80400	73440	76320	71544	6233	8.7
1.25	71520	65040	61800	57960	61344	57048	76800	77088	77040	81120	68676	9031	13.1
0.25	66570	75930	75300	75120	74520	72480	69630	68280	76200	82800	73683	4685	6.4
0.05	70100	65700	84950	82000	88850	77250	65650	62950	69400	78300	74515	8993	12.1
0.01	77700	69755	67165	75845	71610	69615	81305	66150	76300	70280	72573	4950	6.8
0	78430	72875	71280	81840	89540	62975	72600	83930	64185	76395	75405	8404	11.1
g-chain													
gDNA (ng)	Rep1	Rep2	Rep3	Rep4	Rep5	Rep6	Rep7	Rep8	Rep9	Rep10	Mean	SD	CV%
400	123360	127680	121680	126000	122640	121680	123120	120000	124320	121440	123192	2297	1.9
200	142560	140640	134640	135360	123840	126000	121440	122640	121920	120240	128928	8509	6.6
100	118080	108720	112560	111360	115200	114240	118080	108720	122880	126720	115656	5896	5.1
50	120240	115920	119520	115920	119760	126480	124320	107760	117120	120720	118776	5142	4.3
10	110400	108720	118560	112080	103680	107760	116640	106560	122880	121920	112920	6692	5.9
5	102480	121680	109680	115200	119760	114240	103680	97920	109440	119280	111336	8082	7.3
2.5	123600	97200	115920	127440	98640	101760	109200	115920	119520	121440	113064	10792	9.5
1.25	97680	103680	101040	95520	100080	97920	114000	110640	125520	123840	106992	10977	10.3
0.25	100500	110400	103800	102900	101700	105300	98100	98400	121500	115800	105840	7741	7.3
0.05	101880	93015	112725	111600	96795	107550	108450	102600	102420	107010	104405	6283	6.0
0.01	98100	96000	118500	92400	95610	86400	90240	112800	89400	93690	97314	10344	10.6
0	118900	109900	126650	110900	100150	100450	108000	99900	117250	100500	109260	9362	8.6